

A Project Report on

“Online fast and secure digital shopping management system”

*Submitted in the partial fulfillment of the requirements for the
award of the Degree of*

Bachelor of Computer Application

Semester - VI (2025-26)

Submitted by

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Submitted to



Faculty of Science

Department of Computer Science

Atmiya University



ATMIYA UNIVERSITY

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Project ID: BCAC5284

Date: 04-04-2026

CERTIFICATE

This is to certify that,

Mr. / Ms. / Mrs. Dabhi Ramji Rajeshbhai of

Bachelor of Computer Application

Semester VI

Has satisfactorily completed the project on

Online fast and secure digital shopping management system

For, Department of Computer Science,

Faculty of Science, ATMIYA University, Rajkot, Gujarat (India).

Signature
(Project Guide)

Signature
(Head of the Department)

DECLARATION

I hereby declare that the project report titled “**Online fast and secure digital shopping management system**” is my original work and has been completed under the guidance of **Mr. Rahul Bagda**. This project is submitted in the partial fulfillment of the requirements for the award of the Degree of Bachelor of Computer Application, Semester – VI (2025–26) under the Faculty of Science, Department of Computer Science, Atmiya University, Rajkot, Gujarat-India.

I also declare that this work has not been submitted to any other university or institution for any degree, diploma, or certificate.

Name of Student: Dabhi Ramji R.

Enrollment No.: 230801115

Signature:

Date:

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ABSTRACT

In the rapidly evolving landscape of digital commerce, the demand for specialized, high-performance e-commerce platforms is at an all-time high. “Moonchild” is a futuristic, niche e-commerce application designed to cater to the market of high-end consumer electronics and next-generation gadgets. While

generic e-commerce platforms offer broad solutions, they often lack the specialized user experience and inventory management features required for high-value tech products

This project addresses the need for a seamless, secure, and visually immersive online shopping experience. Developed using PHP for server-side logic and MySQL for robust data management, the system ensures data integrity, fast transaction processing, and a responsive user interface built with HTML5, CSS3, and JavaScript.

The core functionality of the system encompasses a comprehensive product management module, a dynamic shopping cart, a secure user authentication system, and an administrative dashboard for real-time inventory and order tracking. The project successfully demonstrates the implementation of a full-stack web application, adhering to modern software engineering principles. The outcome is a fully functional, scalable e-commerce solution that bridges the gap between futuristic product offerings and present-day consumer accessibility.

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Chapter 1

INTRODUCTION

1.1 Overview

The digital revolution has fundamentally transformed how businesses operate and how consumers purchase goods. E-commerce has emerged not just as an alternative to physical retail, but as a primary driver of the global economy. “Moonchild” is a web-based e-commerce application developed to provide a sophisticated platform for buying and selling futuristic electronic products. Unlike traditional online stores, this project focuses on a streamlined, high-performance user experience tailored for tech enthusiasts.

The application serves two primary user roles: the Administrator and the Customer. The Administrator has complete control over the system, including product management, category classification, and order processing. The Customer, on the other hand, enjoys an intuitive interface to browse products, manage their wishlist and cart, and securely place orders. The system is built on a robust architecture that ensures scalability, security, and performance.

1.2 Purpose and Scope

The primary purpose of this project is to automate the manual processes involved in managing a niche electronics store. By digitizing inventory management, order processing, and customer relationship management, “Moonchild” aims to reduce operational errors and enhance efficiency.

Scope of the Project: * User Module: Registration, Login, Profile Management, Product Browsing, Search, Cart Management, Wishlist, and Checkout. * Admin Module: Dashboard analytics, Product CRUD (Create, Read, Update, Delete) operations, Category management, Order status updates, and System settings. * Data Management: Secure storage of user data, order history, and product details using a relational database. * Security: Implementation of secure session management, password hashing, and input validation to prevent common web vulnerabilities.

1.3 Background Information

Historically, retail management relied heavily on manual record-keeping and physical storefronts. This approach is fraught with limitations, including geographical restrictions, limited operating hours, and human error in stock tracking. With the advent of the World Wide Web in the 1990s and the subsequent rise of platforms like Amazon and eBay, the retail paradigm shifted.

Technologically, this project builds upon the established LAMP (Linux, Apache, MySQL, PHP) stack architecture, which powers a significant portion of the web. By leveraging modern PHP (Hypertext Preprocessor) for backend logic and MySQL for relational database management, the system provides a cost-effective yet powerful solution for small to medium-scale e-commerce enterprises.

Chapter 2

PROBLEM DEFINITION

2.1 Real-World Problem

In the current retail sector, specifically within the niche market of high-end electronics, verify capability and inventory accuracy are paramount. Many existing small-scale tech retailers still rely on disjointed systems—using spreadsheets for inventory, separate tools for billing, and manual methods for customer tracking. This fragmentation leads to data inconsistency, stock discrepancies, and a poor customer experience.

Furthermore, generic e-commerce platforms (SaaS solutions) often come with bloated features and recurring costs that are unsustainable for specialized startups. They may not offer the specific customization required to showcase detailed technical specifications of futuristic gadgets, such as holographic displays or neural-link devices, effectively.

2.2 Existing Challenges

- **Manual Inventory Errors:** accurate tracking of high-value stock is difficult with manual systems, leading to overstocking or stockouts.
- **Data Redundancy:** maintaining separate records for sales, customers, and inventory results in duplicated and often conflicting data.
- **Lack of Scalability:** excel-based or paper-based systems cannot handle a growing customer base or increased transaction volume.
- **Customer Inconvenience:** without an online presence, customers are restricted to physical store hours and locations.

2.3 Need for a Technical Solution

To overcome these challenges, a centralized, web-based system is essential. A custom-built e-commerce application addresses these issues by:

1. **Centralizing Data:** combining inventory, sales, and user data into a single relational database.
2. **Automating Processes:** automatically updating stock levels upon purchase and calculating totals/taxes.
3. **Enhancing Accessibility:** providing 24/7 access to the store from any location with internet connectivity.
4. **Improving User Experience:** offering advanced search and filtering capabilities to help users find specific technical products quickly.

Chapter 3

OBJECTIVES OF THE STUDY

The primary objective of this project is to design and develop a fully functional e-commerce website that simplifies the buying and selling process. The specific objectives are as follows:

- **To develop a user-friendly interface** that provides a seamless navigation experience for customers browsing high-tech products.
- **To implement a secure authentication system** ensuring the safety of user accounts and personal information through password encryption and session management.
- **To create a dynamic product catalog** that allows the administrator to easily add, update, and categorize products with rich media and technical specifications.
- **To design a robust shopping cart and checkout system** that handles calculations for sub-totals, shipping costs, and discounts accurately in realtime.
- **To provide an administrative dashboard** for comprehensive monitoring of sales performance, order statuses, and inventory levels.
- **To ensure data integrity and consistency** using a normalized MySQL database structure with proper foreign key constraints.

Chapter 4

SYSTEM ANALYSIS

4.1 Existing System

The existing system largely refers to the traditional method of shopping or using basic, non-integrated digital tools. In a manual or semi-automated setup, the shopkeeper physically checks the stock, manually writes receipts, and maintains a ledger for daily sales.

Working Method

Customers visit the physical store, browse items on shelves, and pay via cash or card at a physical counter. Record keeping is done at the end of the day.

Limitations & Risks

- **Time-Consuming:** manual processing of orders takes time.
- **Human Error:** calculation mistakes in billing.
- **No Remote Access:** business stops when the shop closes.
- **Limited Reach:** customer base is limited to the local vicinity

4.2 Proposed System

The “Moonchild” proposed system is a web-based application that automates the entire retail lifecycle. It integrates the frontend store with a backend administration panel.

System Workflow

Users register and log in to the portal. They browse categories (e.g., Laptops, VR Headsets), view detailed product pages, add items to a cart, and proceed to checkout. The system records the order in the database and instantly deducts the stock. Administrators view these orders in real-time and update their status (e.g., “Shipped”, “Delivered”).

Advantages

- **Global Reach:** accessible from anywhere.
- **Automation:** instant stock updates and bill calculation.
- **Data Analysis:** easier to track popular products and revenue.
- **Cost-Effective:** reduces the need for physical retail space and staff.

4.3 Feasibility Study

Technical Feasibility

The project uses PHP and MySQL, which are standard, open-source technologies supported by almost all web servers. The hardware requirements are minimal (client-side browser, server-side hosting), making it technically viable. The development team possesses the necessary skills in HTML, CSS, JS, and SQL.

Economic Feasibility

Since the project utilizes open-source software (XAMPP stack, VS Code), there are no licensing costs involved. The primary cost is development time and potential hosting fees. Compared to purchasing a proprietary enterprise ERP setup, this custom solution is highly consistent with the budget constraints of a startup.

Operational Feasibility

The system is designed with a focus on User Experience (UX). The interface is intuitive, requiring no special training for customers. The admin panel is structured logically, ensuring that the store manager can operate it with basic computer literacy. Therefore, operational adoption is expected to be smooth.

4.4 Requirement Specification

Hardware Requirements

Component	Minimum Specification	Recommended Specification
Processor	Intel Core i3 or equivalent	Intel Core i5 or higher
RAM	4 GB	8 GB or higher
Storage	256 GB HDD/SSD	512 GB SSD
Monitor	1366x768 resolution	1920x1080 resolution
Internet	5 Mbps connection	Broadband / Fiber connection

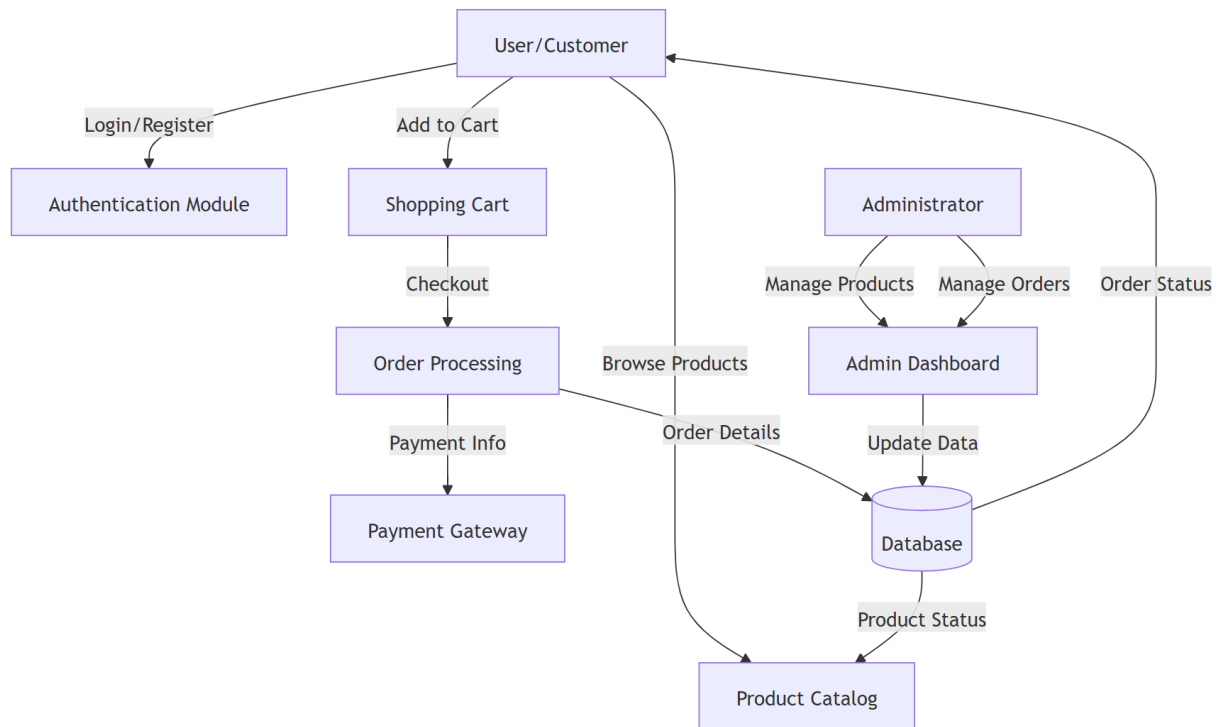
Software Requirements

Component	Requirement
Operating System	Windows 10/11, Linux, or macOS
Server Stack	XAMPP / WAMP / LAMP (Apache, MariaDB/MySQL, PHP)
Frontend Technologies	HTML5, CSS3, JavaScript (ES6)
Backend Language	PHP 8.0+
Database	MySQL 5.7+
Web Browser	Google Chrome, Firefox, or Edge
Code Editor	Visual Studio Code

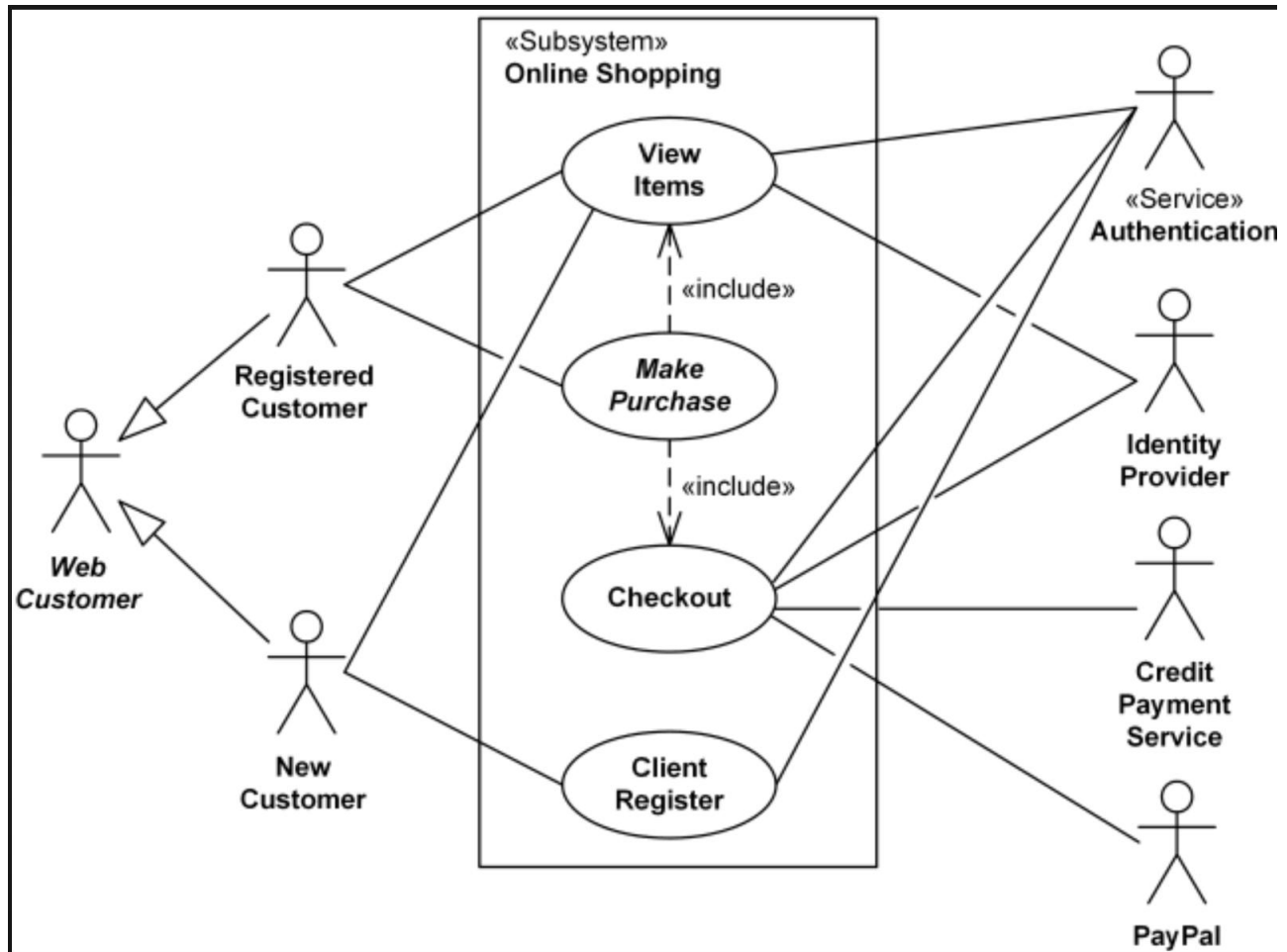
Chapter 5

SYSTEM DESIGN

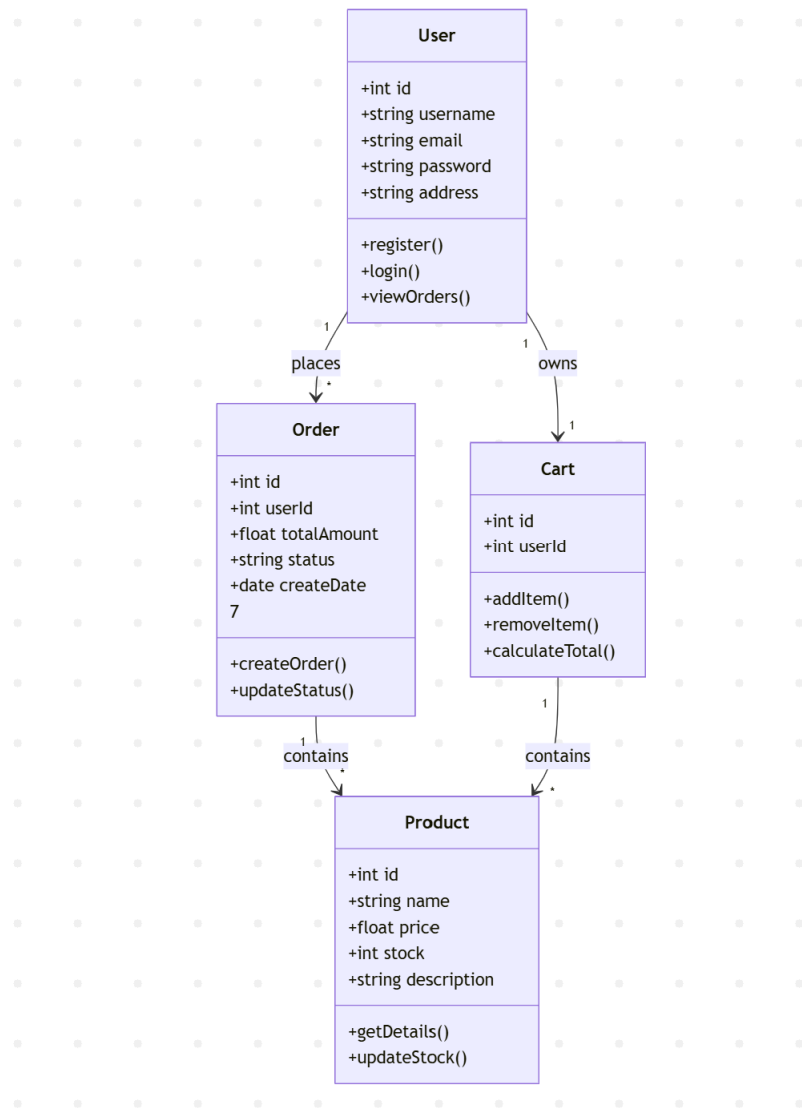
5.1 Data Flow Diagram (DFD)



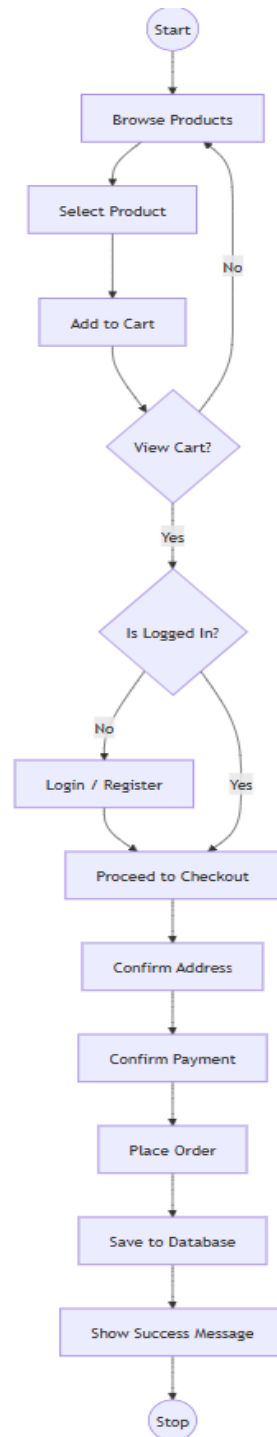
5.2 Use Case Diagram



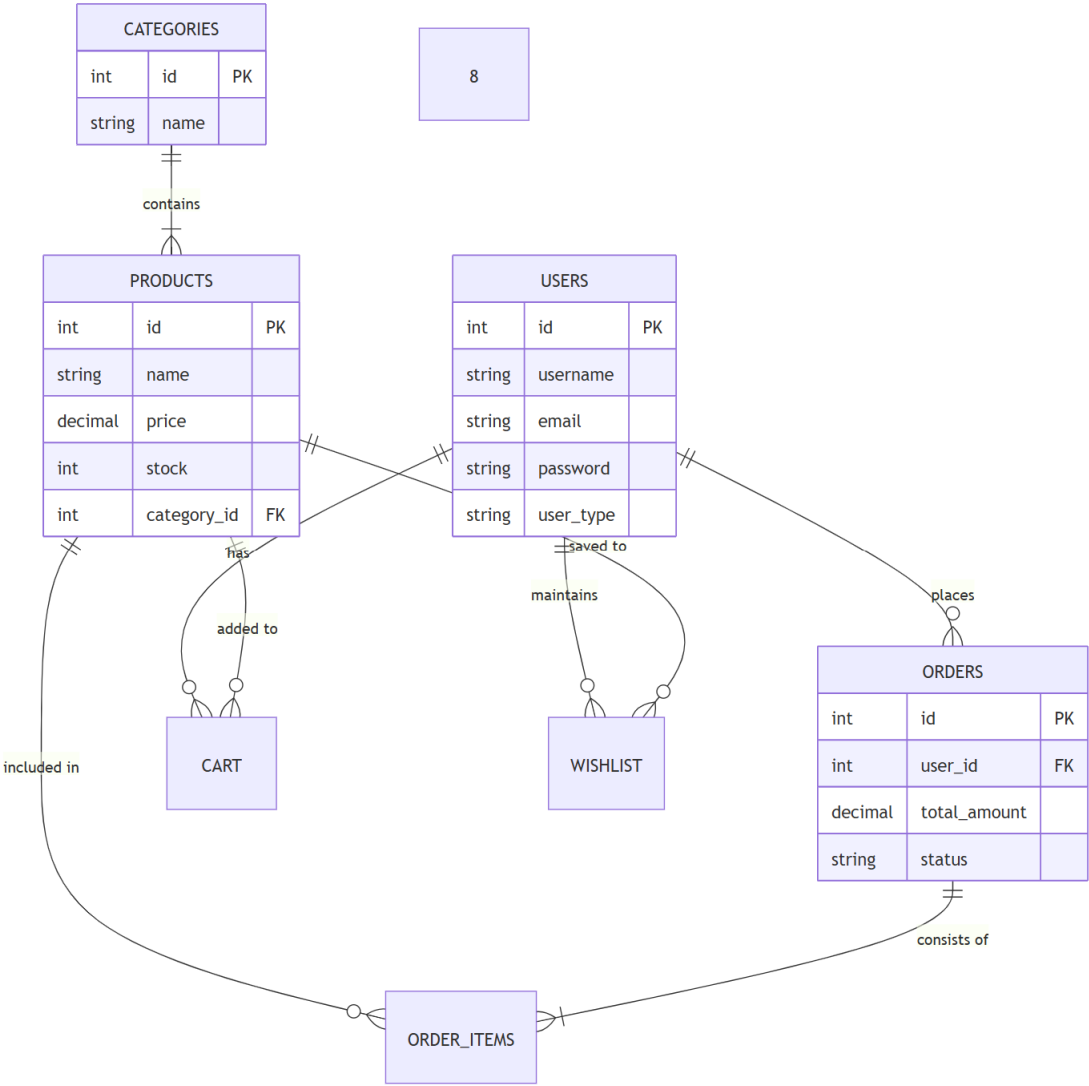
5.3 Class Diagram



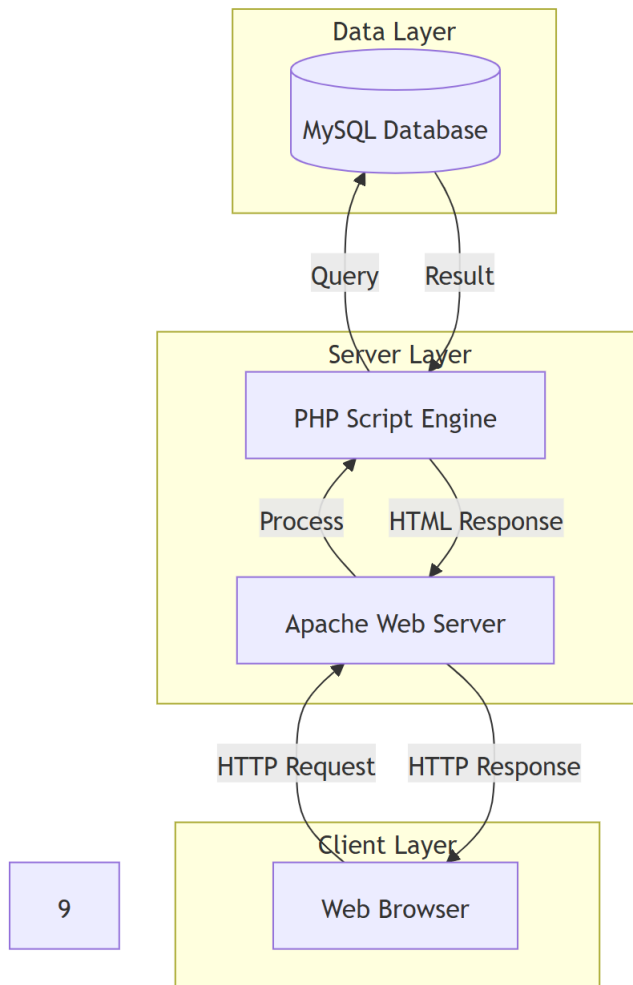
5.4 Activity Diagram (Order Process)



5.5 Entity Relationship (ER) Diagram



5.6 System Architecture



Chapter 6

IMPLEMENTATION

6.1 Module Description

User Authentication Module

This module handles user security. It uses PHP Sessions (session_start()) to track user state. Passwords are hashed before storage using standard algorithms to ensure security. * **Input:** Username, Email, Password. * **Process:** Validation, Hashing, Database Insertion/Lookup. * **Output:** Session creation and redirection to Dashboard.

Product Management Module

Allows the admin to manage the inventory. Products are categorized (e.g., Electronics, Laptops) for easy filtering. The products table in the database stores all implementation details including image paths and stock levels. * **Technologies:** PHP Forms, File Upload for images.

Shopping Cart Module

A crucial component that allows users to accumulate items before purchase. The cart is persistent for logged-in users, stored in the cart database table. * **Process:** Checks current stock, adds item to cart table, updates quantity if item already exists.

Order Processing Module

Handles the final transaction. It calculates the total price, including shipping and taxes. When an order is placed, data moves from the cart table to the orders and order_items tables, and the product stock is decremented to prevent overselling.

6.2 Technologies Used

- **HTML5 (HyperText Markup Language):** Used for the structural skeleton of the web pages.
- **CSS3 (Cascading Style Sheets):** Used for styling the application, including layout, colors, and fonts (using shop.css and style.css).
- **JavaScript:** Used for client-side interactivity, form validation, and dynamic content updates (script.js).
- 10• **PHP (Hypertext Preprocessor):** The core server-side scripting language used to handle business logic and database connectivity.
- **MySQL:** The Relational Database Management System (RDBMS) used to store all application data.
- **Apache:** The web server software responsible for serving the application files.

6.3 Development Methodology

The project followed the **Waterfall Model** of software development. This linear approach ensured that each phase—Requirements Analysis, System Design, Implementation, Testing, and Deployment—was completed and verified before moving to the next. This structure was ideal for a university project with clear, fixed requirements.

1. **Requirement Gathering:** Identified the need for a niche tech store.
2. **Design:** Created ER diagrams and schema.
3. **Coding:** Developed frontend and backend modules.
4. **Testing:** Verified detailed functionality

Chapter 7

TESTING

Software testing is a critical phase to ensure the system is defect-free and meets the requirements.

7.1 Types of Testing Performed

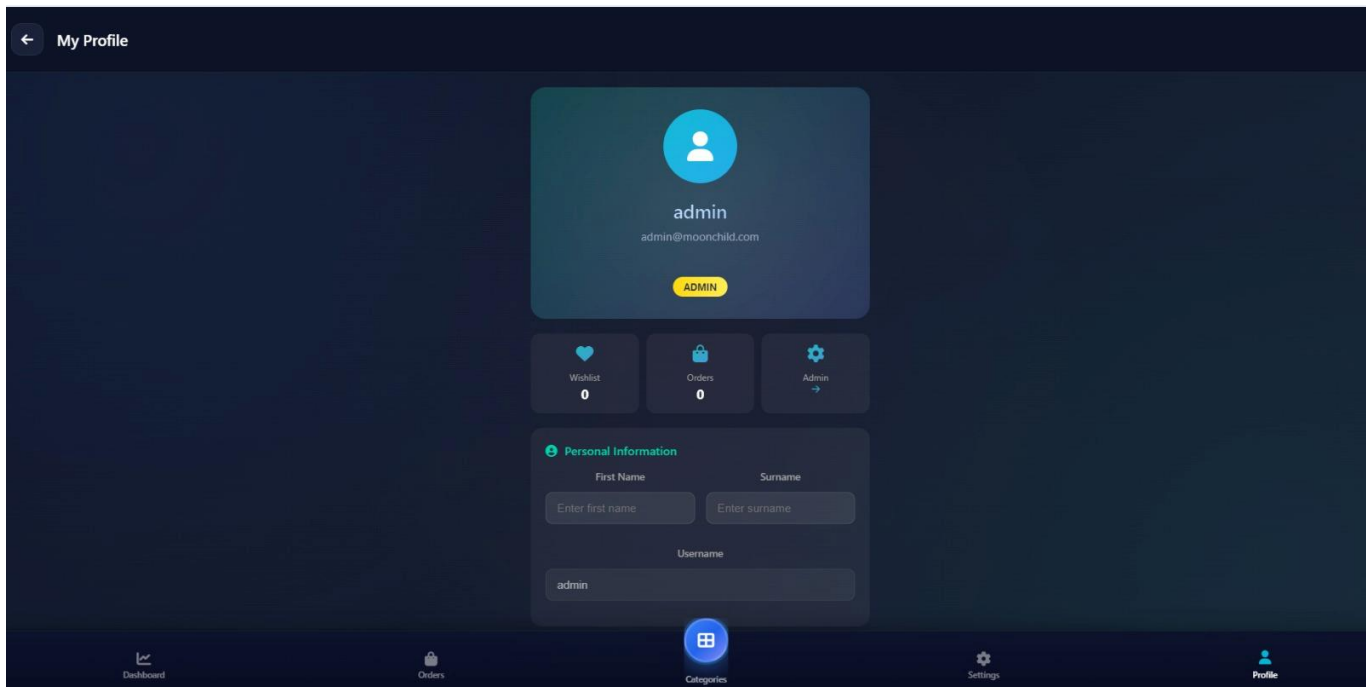
- **Unit Testing:** Individual modules (e.g., Login function, Cart calculation) were tested in isolation to ensure logic correctness.
- **Integration Testing:** Verified that modules work together, for example, ensuring that adding a product updates the cart total correctly.
- **System Testing:** The complete application was tested as a whole to validate end-to-end flows (from Registration to Order Placement).
- **Validation Testing:** Input forms were tested to ensure they reject invalid data (e.g., negative prices, invalid email formats).

7.2 Test Cases

TC ID	Test Case	Steps	Expected Result	Actual Result	Status
TC 01	User Registrstion	Enter valid details and submit	Account created, redirected to login	As Expected	Pass
TC 02	User Login	Enter valid credentials	Login successful, session started	As Expected	Pass
TC 03	Invalid Login	Enter wrong password	Show error message	Error displayed	Pass
TC 04	Add to Cart	Click 'Add to Cart' on product	Item added cart count +1	As Expected	Pass
TC 05	Stock Validation	Try to buy more than stock	System prevents addition	As Expected	Pass
TC 06	Place Order	Proceed to checkout	Order saved, Cart cleared	As Expected	Pass
TC 07	Admin Login	Login with admin credentials	Redirect to Admin Dashboard	As Expected	Pass
TC 08	Add Product	Admin adds new product	Product appears on Homepage	As Expected	Pass

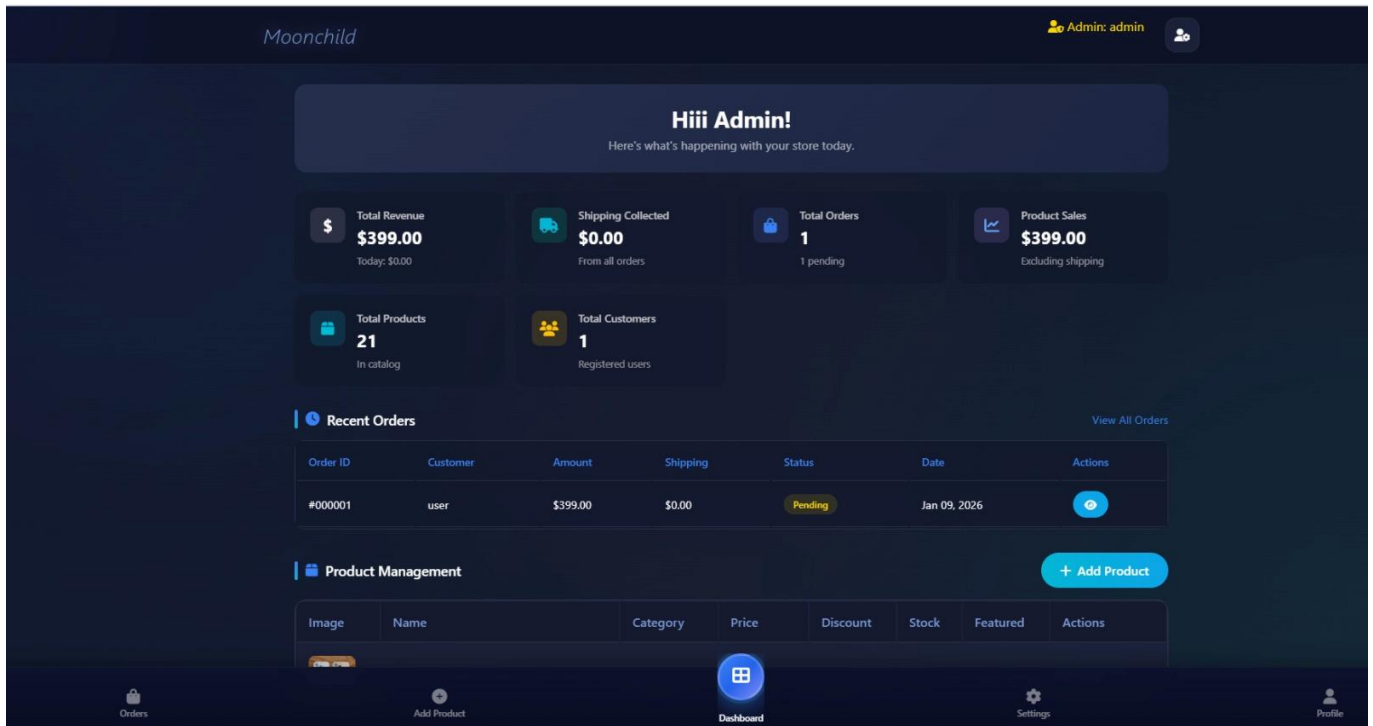
Chapter 8

OUTPUT SCREENSHOTS

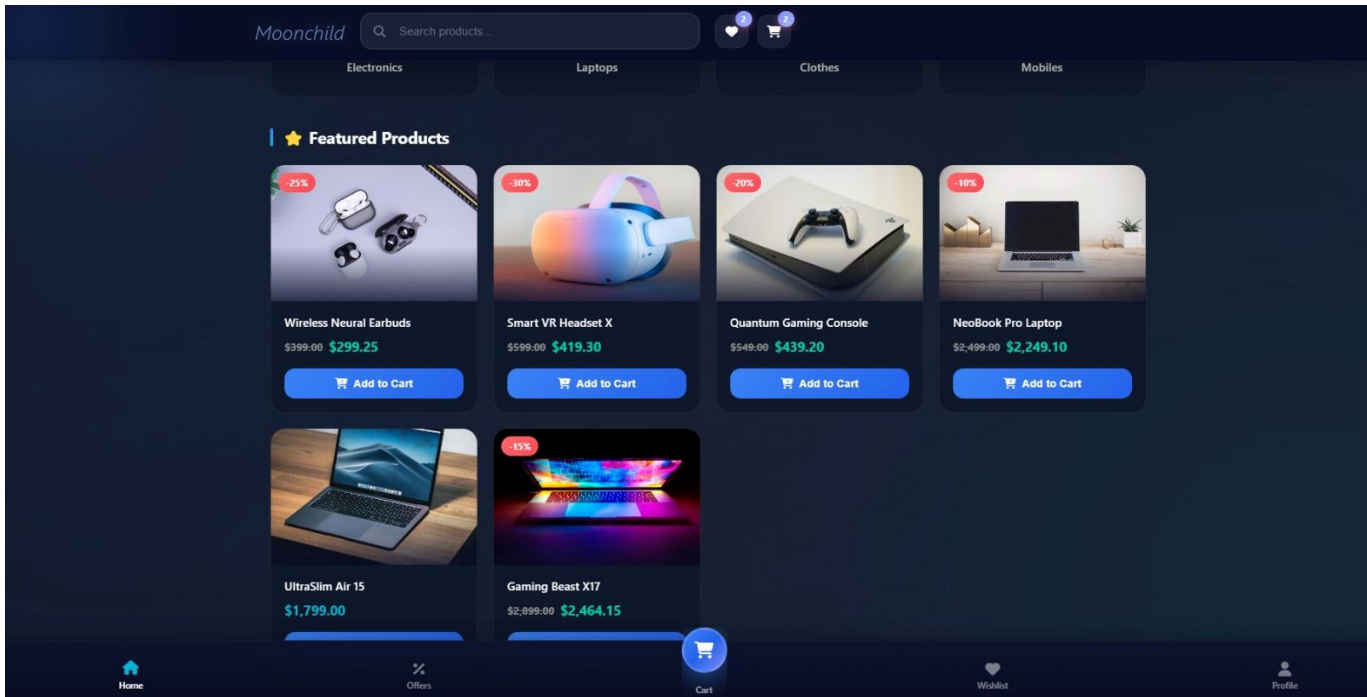


The image shows a dark-themed user profile dashboard. The background has a blue gradient design with soft lighting effects. At the top left, there is a back arrow and the title “My Profile.”

In the center, a profile card displays a circular avatar icon, the username “admin,” and the email “admin@moonchild.com.” A yellow badge labeled “ADMIN” indicates the user’s role.



At the top, the brand name “Moonchild” appears on the left, and “Admin: admin” is shown on the right with a user icon. In the center, there is a welcome message: “Hiii Admin! Here’s what’s happening with your store today.”



The image shows a dark-themed e-commerce homepage for a store named “Moonchild.”

At the top, there is the store name on the left, a search bar in the center, and wishlist and cart icons on the right with notification badges. Below this, product categories such as Electronics, Laptops, Clothes, and Mobiles are displayed.

The main section is titled “Featured Products.” Several product cards are shown with images, discount badges, product names, original and discounted prices, and “Add to Cart” buttons.

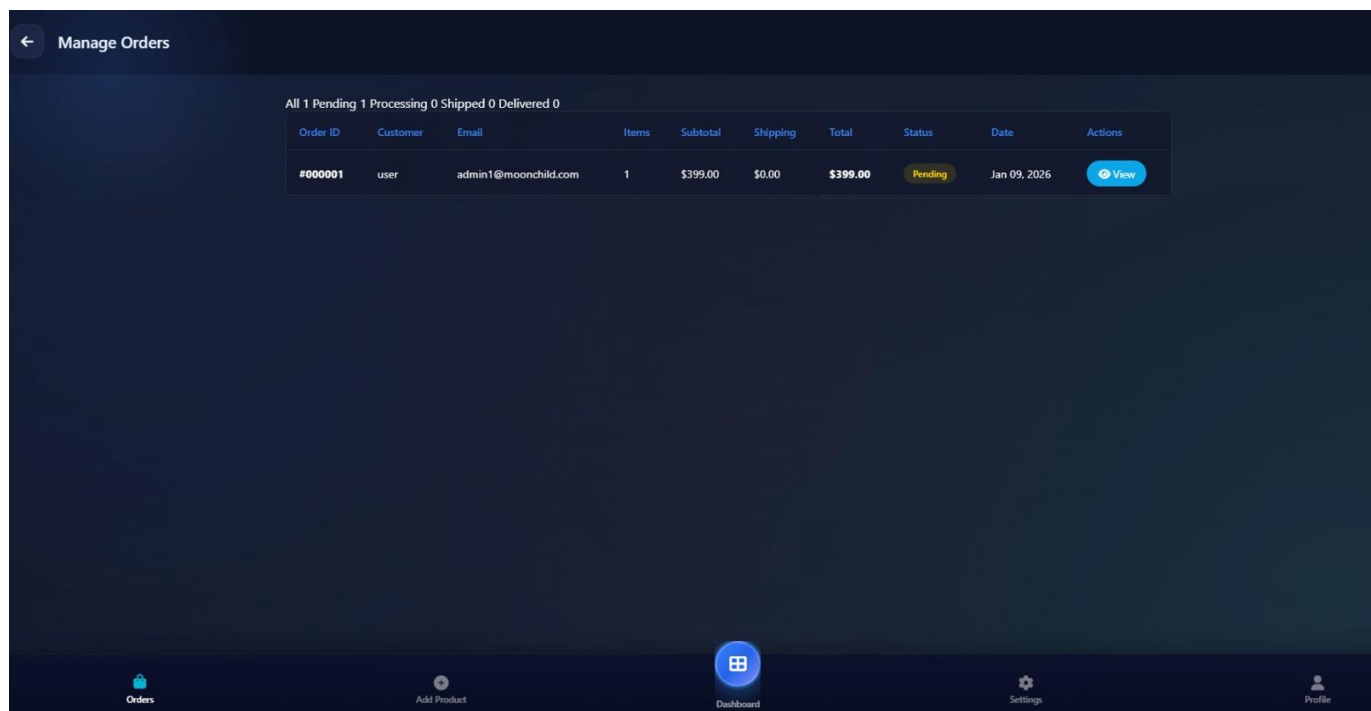
The image shows a dark-themed "Add Product" page from an admin dashboard. At the top left, there is a back arrow and the title "Add Product." In the center, a product form is displayed inside a rounded card. The form includes the following fields:

- Product Name *
- Category * (with a dropdown menu showing "Select Category")
- Price (\$) *
- Discount (%) - Leave 0 for no discount (with a value of 0)
- Stock Quantity * (with a value of 0)
- Brand
- Image URL *
- Description

At the bottom, there is a navigation bar with icons for Orders, Add Product, Dashboard, Settings, and Profile.

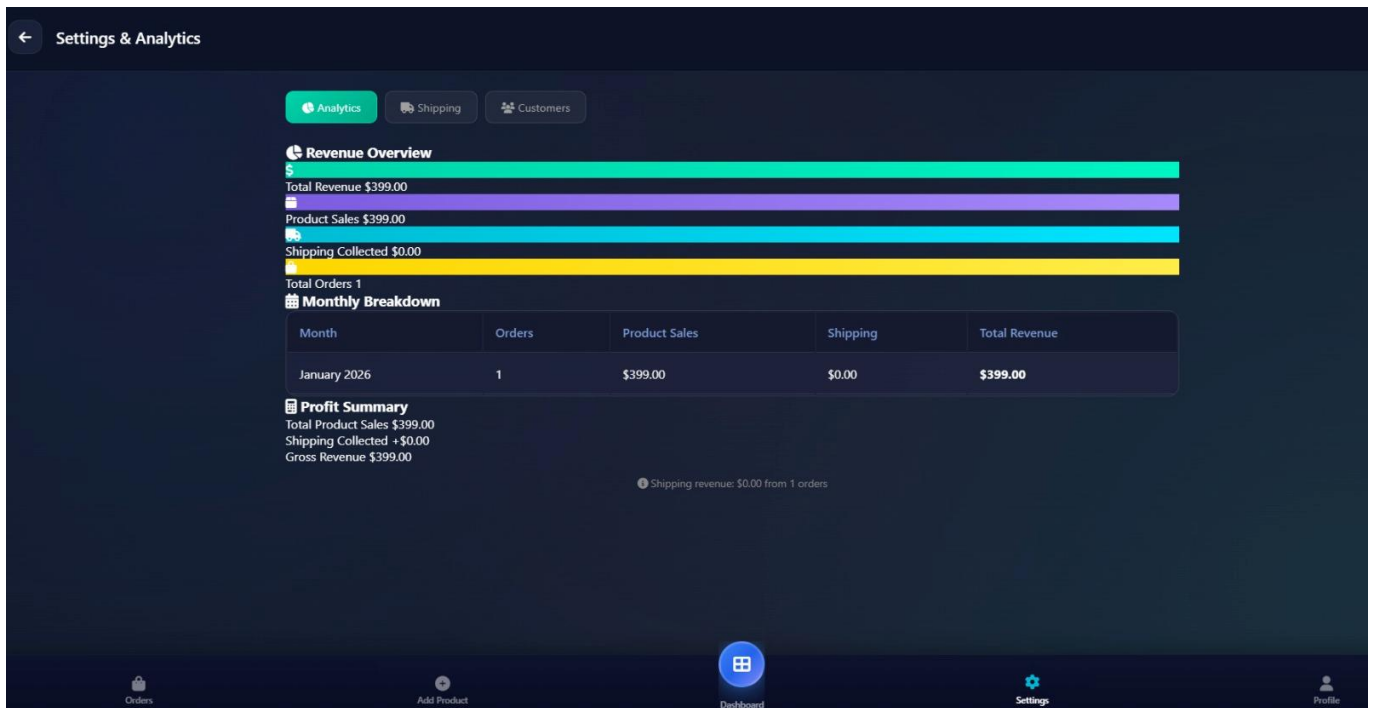
The image shows a dark-themed “Add Product” page from an admin dashboard.

At the top left, there is a back arrow and the title “Add Product.” In the center, a product form is displayed inside a rounded card.



The image shows a dark-themed “Manage Orders” page from an admin dashboard.

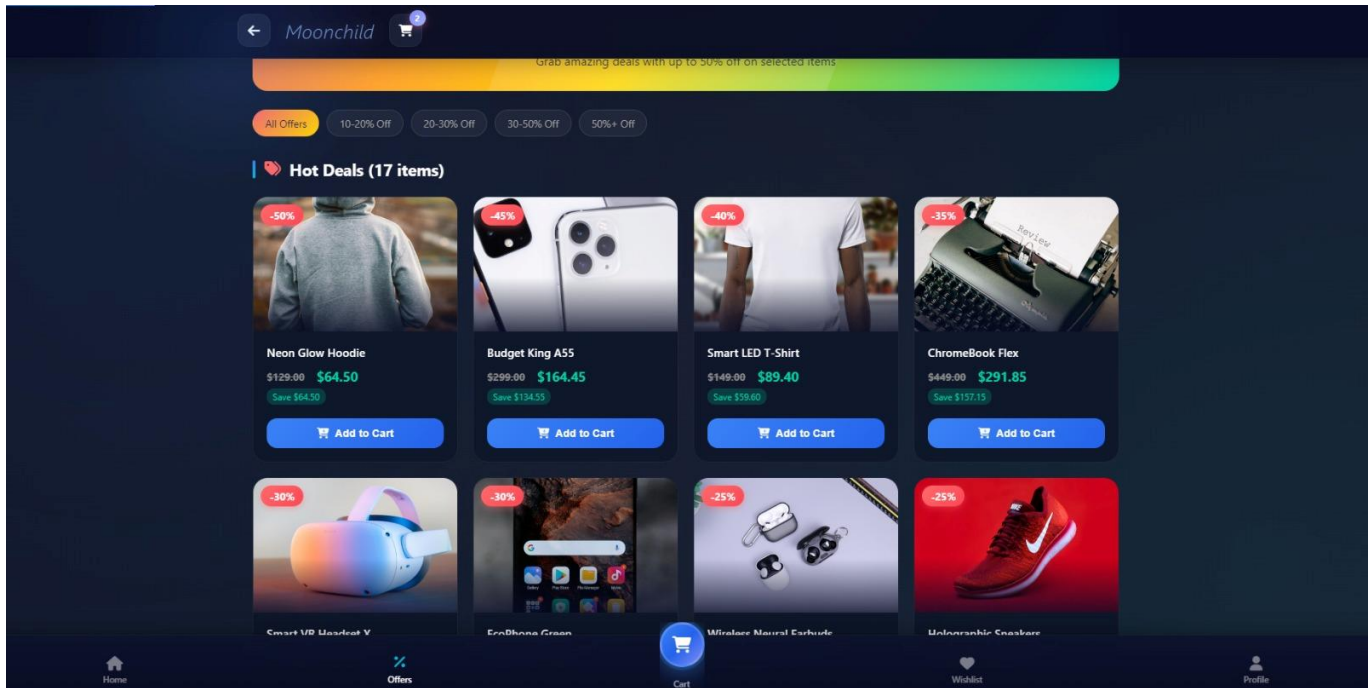
At the top left, there is a back arrow and the title “Manage Orders.” Below it, a summary line displays order counts: All 1, Pending 1, Processing 0, Shipped 0, Delivered 0.



The image shows a dark-themed **Settings & Analytics** dashboard screen.

At the top left, there is a back arrow and the title “**Settings & Analytics.**” Below it are three tabs:

- **Analytics** (selected/highlighted)
- Shipping
- Customers

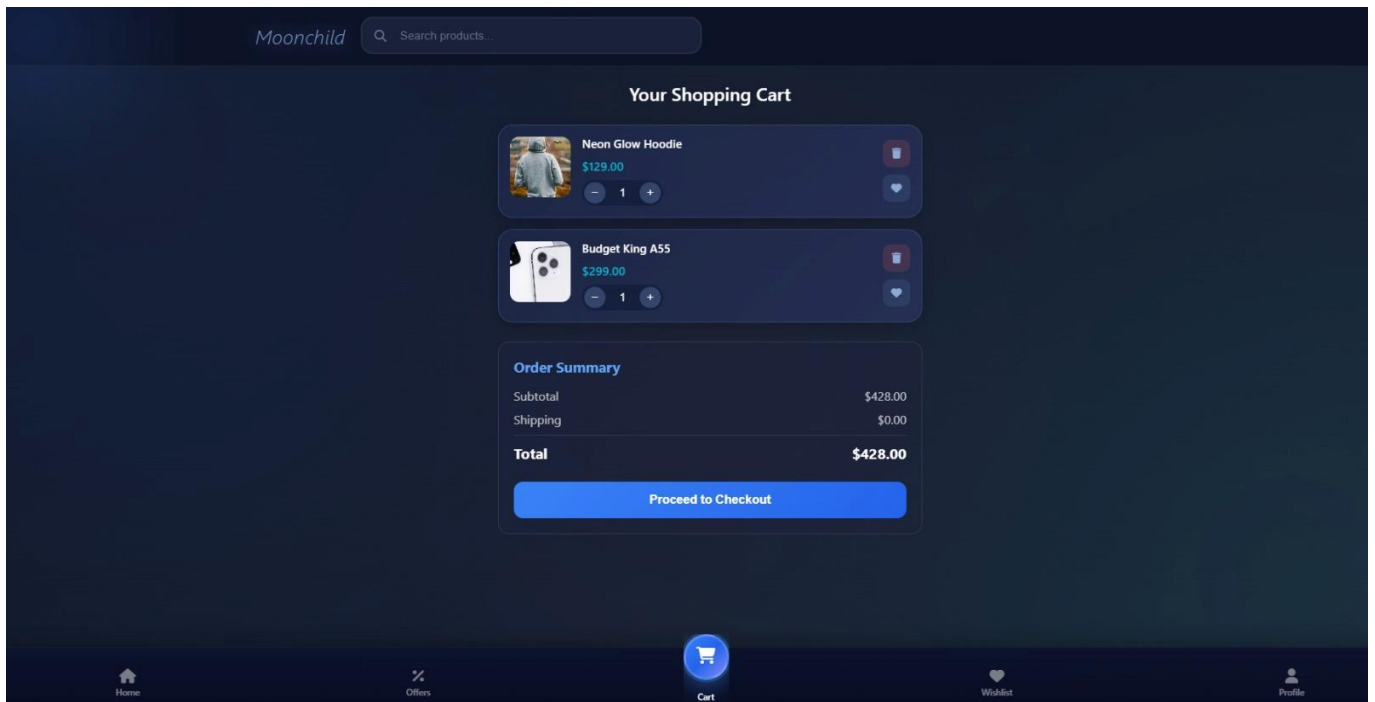


The image shows a dark-themed **Moonchild** shopping app screen displaying discounted products.

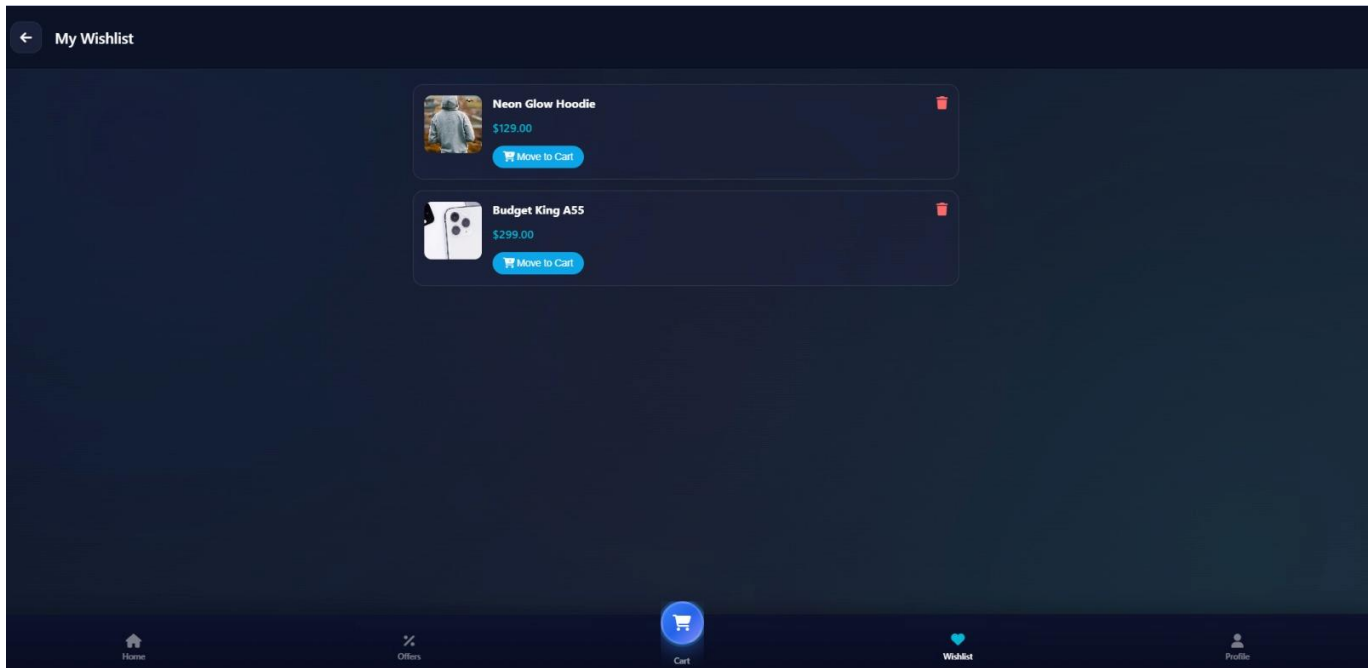
At the top:

- A back arrow on the left
- The store name “**Moonchild**” in the center
- A shopping cart icon on the right with a small badge showing **2 items**

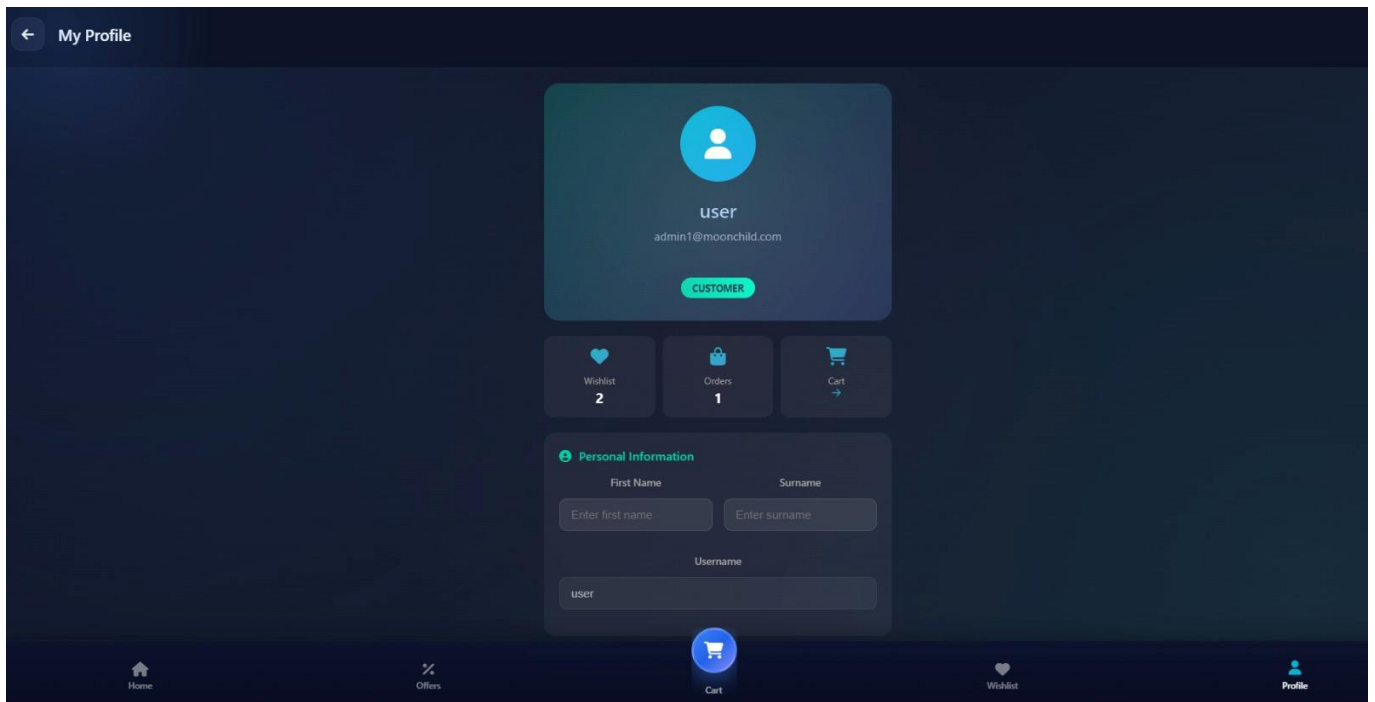
Below is a colorful banner promoting deals (up to 50% off selected items).



he image shows a **shopping cart user interface** for an e-commerce platform named **Moonchild**. The design features a dark theme with a sleek, modern aesthetic.



The image displays a **wishlist user interface** for an e-commerce application, maintaining the dark-themed aesthetic of the previous screen.



These images showcase the **Wishlist** and **Profile** interfaces for the Moonchild application, maintaining a consistent dark aesthetic.

Chapter 9

CONCLUSION

The “Moonchild” E-Commerce System has been successfully designed, developed, and tested. The project achieves its primary objective of providing a robust, user-friendly platform for buying and selling specialized electronic products.

Key Achievements:

- * Successfully implemented a secure and scalable architecture using the PHP/MySQL stack.
- * Created a seamless user flow from registration to checkout.
- * Developed a comprehensive admin panel that simplifies inventory management.
- * Bridged the gap identified in the problem statement by providing a dedicated platform for high-tech gadgets.

Limitations:

While the system is functional, it currently relies on a “Cash on

Delivery” or simulated payment method. Real-time integration with banking APIs is simulated for security reasons in this academic scope. Additionally, the recommendation engine is basic, relying on category matching rather than machine learning algorithms.

In conclusion, this project serves as a solid foundation for a scalable e-commerce business. It demonstrates the practical application of software engineering concepts and web development technologies to solve real-world retail challenges.

Chapter 10

FUTURE SCOPE

The current system lays the groundwork for a sophisticated e-commerce platform. Future enhancements can elevate “Moonchild” to an enterprise-level application:

1. **AI-Powered Recommendations:**
integrate Machine Learning algorithms to analyze user behavior and suggest products based on browsing history and purchase patterns.
2. **Payment Gateway Integration:**
implement secure, real-time payment processing using Stripe or PayPal APIs, and potentially cryptocurrency payments to match the futuristic theme.
3. **Mobile Application:**
develop a React Native or Flutter mobile app to tap into the mobile-commerce market.
4. **Augmented Reality (AR):**
given the nature of the products (e.g., VR headsets, furniture), an AR feature allowing users to visualize products in their space would be a valuable addition.
5. **Chatbot Support:** integrate an AI chatbot for 24/7 customer support and order tracking inquiries.

Chapter 11

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Chapter 12

APPENDIX

A. Sample Database Configuration

```
-- Database Name: moonchild_db  
-- User: root  
-- Host: localhost
```

```
CREATE DATABASE moonchild_db;  
USE moonchild_db;
```

B. Abbreviations

- **PHP:** Hypertext Preprocessor
- **SQL:** Structured Query Language
- **HTML:** HyperText Markup Language
- **CSS:** Cascading Style Sheets
- **JS:** JavaScript
- **RDBMS:** Relational Database Management System
- **IDE:** Integrated Development Environment
- **XAMPP:** Cross-Platform, Apache, MySQL, PHP, Perl

C. Acknowledgement of Tools

The development of this project utilized various open-source tools including Visual Studio Code, XAMPP Server, and Google Chrome Developer Tools.